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Inflammatory blood parameters as prognostic factors for surgical site infection after primary hip or knee arthroplasty: a systematic review protocol

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Inflammatory blood parameters as prognostic factors for surgical site infection after primary hip or knee arthroplasty: a systematic review protocol.

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Abstract

Introduction

Surgical site infection (SSI) is a potential complication of surgical procedure. In orthopaedics, the SSI after implant surgery is a disaster both for patients and surgeons. Although predictive tools for SSI, such as ACS calculator, or NHSN risk index are available, none of them estimate early infection based on inflammatory blood parameters. The inflammatory process can be measured using several parameters including interleukin-6, c-reactive protein, neutrophil to lymphocyte ratio, white blood cell count, erythrocyte sedimentation rate, or procalcitonin. This systematic review aims to determine whether inflammatory blood parameters could be used as significant predictive factors for SSI after primary hip or knee arthroplasty.

Methods and analysis

A systematic review of randomized controlled trials, cross-sectional studies, case-control studies, and cohort studies, published in English, will be searched in the following electronic bibliographic databases: MEDLINE, Embase, PubMed, Cochrane Central Register of Controlled Trials, and Web of Science. Studies performed in adult patients of all ages who underwent knee or hip arthroplasty, studies containing data on the risk/prognostic factors for pre or post knee or hip arthroplasty SSI and studies with a minimum follow up of 30 days after surgery will be included. A standardized form will be used to extract data from the included studies comprising study characteristics, participant characteristics, details of the intervention, study methodology and outcomes. QUADAS-2 and STARD checklist will be used to assess risk of bias. Heterogeneity will be assessed using Cochran χ^2 statistic and I^2 statistics where applicable. Grading of Recommendations Assessment, Development and Evaluation and Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidance will be used to report findings.

Ethics and dissemination

No ethics approval is required. The findings will be disseminated at national and international scientific sessions, also to be published in a peer-reviewed journal.

PROSPERO registration number

CRD42020147925

Keywords

Adult orthopaedics, hip, knee, infection control

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Strengths and limitations of this study

This protocol reduces the risk of duplication and biases, makes methods and processes that will be used transparent and enables peer reviewing.

The protocol offers the highest level of evidence for informed decisions based on this systematic review.

To the best of our knowledge, this will be the first systematic review assessing the prognostic value of routine screening of inflammatory blood parameters for SSI in primary hip or knee arthroplasty.

The scarcity of studies focused on patients after primary knee or hip arthroplasty containing data on the risk/prognostic factors for SSI with a minimum follow up of 30 days, an expected heterogeneity in design, varying methodological quality across studies and limitation of review search to English language studies may hinder meta-analysis and interpretation of findings.

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Introduction

Surgical site infection (SSI) is a potential complication following surgical procedure.[1] The presence of SSI leads to an increase both in morbidity and in mortality rates, thus contributing to higher healthcare costs.[2,3] In orthopaedics, SSI after implant surgery is a disaster both for the patients and surgeons.[4] SSI prevalence may be reduced by various preoperative and postoperative measures. One of the measurements in which SSI is affected is antibiotic prophylaxis.[5,6] It is desirable to adjust antibiotic prophylaxis length for arthroplasty, according to individual SSI risk estimated by appropriate processes. Although predictive tools for SSI, such as ACS (*American College of Surgeons*) surgical risk calculator, or NHSN (*The National Healthcare Safety Network*) risk index are available, none of them estimate early infection based on inflammatory blood parameters (IBD).[8] In the acute-phase, cells of the immune system migrate to the site of injury in a carefully orchestrated sequence of events mediated by cytokines and acute-phase proteins.[9] The inflammatory process can be measured applying one of the several of biochemical markers, including interleukin-6 (IL-6), c-reactive protein (CRP), neutrophil to lymphocyte ratio (NLR), white blood cell count (WBC), erythrocyte sedimentation rate (ESR) or procalcitonin (PCT). NLR is an easy, cheap, widely used, robust and convenient biomarker during inflammation, and has been shown to predict outcomes in cardiovascular diseases.[10,11] After a surgical trauma neutrophil blood levels via cytokines are increased, and the lymphocyte count is decreased. ESR is the rate how quickly red blood cells settle at the bottom of test tube with blood sample. In case of disease, high ESR is commonly due to an increased protein level in the blood, such as the increase in acute-phase proteins in inflammatory diseases. The ESR is nonspecific and, therefore, of little diagnostic value, but serial tests enable to more efficiently follow the progress of the disease. CRP, also known as nonspecific acute-phase response marker, is instigated by tissue damage, infection, inflammation, and malignancy. CRP is under control by IL-6 originating at the site of pathology. Serum concentration starts to rise within 6 hours, peaking in around 48 hours, from the time the given pathology occurs. CRP is still the most widely used biomarker for diagnosing postoperative infections.[11,12] PCT is one of the several bloodstream biomarkers investigated as an early predictor of sepsis. It is produced rapidly in response to bacterial stimuli but not in systematic inflammatory response syndrome of non-infectious origin.[11] IL-6 is one of the polypeptides secreted by cells triggered in response to injury, in the broad category of small proteins cytokines. The role of cytokines is to organize a concerted response to tissue trauma, involving cells recruited to the injury site. IL-6 is classified as the inflammation cytokine of acute-phase. To better understand the clinically significant responses to surgical trauma, it is important to elucidate the underlying mechanisms that regulate the inflammatory response to this type of tissue injury. The native immune system recognizes bacteria through pattern recognition receptors, such as toll-like receptors. These receptors stimulate, among others, the secretion of IL-6, which is also stimulated by interleukin-1 (α/β) and the tumour necrosis factor (α/β).[10] The present systematic review will primarily assess the response of IBP to trauma caused by orthopaedic surgery, and evaluate the clinical utility of quantitative measurements of IBP as predictive factors for infection. To our knowledge, there is no published systematic review addressing the same question. Using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA-P) checklist[13], we propose a systematic and reproducible strategy to search the available literature.

Methods and analysis

The search strategy will use tools that improve methodological transparency, reproducibility of the results and evidence synthesis. To ensure the systematic search of available literature, we used the Population, Intervention, Comparison, Outcome and Study design (PICOS) strategy. Following the PICOS strategy, we elaborated the guiding question of this review: *Can haematologic parameters (IL-*

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6, CRP, NLR, ESR, WBC, PCT) be effective as predictive factors for SSI in knee or hip arthroplasty? The International Prospective Register of Systematic Reviews registration number is CRD42020147925. The planned period of this review study is from 1 September 2019 to 30 January 2021.

Inclusion criteria for study selection

The participants, interventions, comparators and outcomes, as well as the type of studies were defined according to PICOS strategy.

Types of studies

Cross-sectional, cohort, case-control, randomized controlled trial study designs will be included, limited to publications in English language. Case reports will be excluded.

Types of participants

Studies with adult patients of all ages who underwent primary hip or knee arthroplasty will be included. The minimum follow-up after the operation must be 30 days. Patients with clinical signs of infection, neoplasia, inflammatory diseases (rheumatoid arthritis, Crohn, HIV, etc.) will be excluded.

Types of interventions or exposures

IBP determination such as IL-6, CRP, WBC, ESR, NLR and PCT.

Types of comparators

We will compare the prognostic value of IBP to each other.

Types of outcomes measures

The outcome measures will be the IBP that can be used as significant predictive factors for surgical site infection development.

Search methods for the identification of studies

Electronic searches

We will conduct this systematic review and meta-analysis in accordance with the PRISMA guidelines.[14] The following electronic bibliographic databases will be searched: PubMed (MEDLINE), Cochrane Central Register of Controlled Trials (Cochrane databases), Web of Science. The search terms will include both medical subject headings (MeSH) and keywords related to the topics about IBP or risk factors, or prosthesis-related infection or surgical site infection and knee or hip arthroplasty. The sample of the search strategy for MEDLINE is presented in Table 1. Similar search strategy will also be utilized to any other electronic database.

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Table 1 Search strategy for MEDLINE database

Number	Search terms
1	prosthesis joint infection or prosthesis related infection
2	prosthetic infection or inflammation or acute-phase reaction or seroma or abscess or systematic inflammatory response syndrome or sepsis
3	wound infection
4	surgical site infection or surgical wound infection
5	arthroplasty or arthroplasty, replacement or arthroplasty replacement, hip or arthroplasty replacement, knee
6	joint replacement
7	joint prosthesis or hip prosthesis or knee prosthesis or metal-on-metal join prostheses
8	joint arthroplasty
9	risk factor or risk factors
10	predictor
11	inflammatory marker
12	biological markers or hematologic tests
13	leukocyte count
14	blood sedimentation rate
15	neutrophils
16	lymphocytes
17	neutrophil to lymphocyte ratio
18	interleukin 6
19	c reactive protein
20	procalcitonin
21	1 or 2 or 3 or 4
22	5 or 6 or 7 or 8
23	9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20
24	21 and 22 and 23
25	limit to 24 (english language and yr="1902-Current"

Note: For purpose of this table, search terms are presented without additional settings like “focus, explode, multi-purpose”, etc.

Searching other resources

Manual searches will be performed mainly to retrieve dissertations, ongoing trials, internal reports etc, in order not to miss any relevant studies. Additionally, we will search the bibliographies of included publications manually for further, potentially eligible papers.

Data collection and analysis

Selection of studies

All electronically and manually searched records will be merged to remove duplicate citations. Titles, abstracts and full-texts of studies retrieved using the search strategy will be screened independently through the EndNote software by at least two review authors (PD, ARP) to identify studies that potentially meet the inclusion criteria in the two-stage selection process. The initial screening of studies will be based on the information contained in their titles and abstracts. This screening will be conducted by two independent investigators (PD and ARP). In case reviewers disagree, the article will be re-evaluated. If the disagreement persists, it will be resolved by a third reviewer (KML). Full-paper screening will be conducted by the same independent investigators. The reasons for which any

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literature excluded will be recorded. The details of the selection process are shown in the PRISMA flow chart (Fig. 1).

Data extraction and management

A standardized form will be used to extract data from the included studies for study quality assessment (Grading of Recommendations Assessment, Development and Evaluation (GRADE) algorithm will be used)[15] and evidence synthesis. Extracted information will include study characteristics (design, settings, year of publication), participant characteristics (mean age, percentage of each gender), details of intervention (blood inflammatory parameters with all characteristics), study methodology and outcomes (surgical site infection, prosthesis-related infection, risk/prognostic factors, limitations). Missing data will be discussed with the authors of studies, where applicable. Two independent Microsoft Excel spreadsheets will be elaborated for two reviewers (PD and ARP) to summarize the data from the included studies. Then, the spreadsheets will be combined into one. Disagreements will be resolved by a third investigator (KML).

Assessment of risk of bias in included studies.

QUADAS-2 tool[16] and STARD checklist[17] will be used to assess the risk of bias in the included studies by two reviewers (PD and ARP). The disagreement in the risk of bias assessment or the level of evidence (GRADE) will be resolved by consensus.

Dealing with missing data

If the required data are unclear or not reported in any of the articles, the reviewer will contact the author by email, or mail to collect the missing data as completely as possible.

Assessment of heterogeneity

Heterogeneity will be assessed using Cochran χ^2 statistic and I^2 statistics where applicable.

Data synthesis and analysis

We will conduct a narrative synthesis for the findings from the included studies, structured around the type of intervention, target population characteristics, type of outcome and intervention content. One reviewer (PD) will synthesize the data. Three other reviewers will check the results and discuss any discrepancies to reach consensus. If there is substantial clinical homogeneity of the participants and assessment methods, meta-analysis will be conducted to estimate the risk/protectability of each IBP in relation to knee or hip arthroplasty.

Assessment of publication bias

The funnel plot will be used to judge whether a publication bias exists if at least 10 articles are included. We will investigate the (a)symmetry in (inverted) funnel plots which show the association between effect size and the accuracy of the study.

Subgroup analysis

When the assessment suggests significant heterogeneity, we will perform a subgroup analysis based on study-level characteristics including diagnosis of the knee or hip osteoarthritis, type of surgery (based on how invasive method was used), use of antibiotic prophylaxis (with other characteristics), thrombosis prevention, average duration of follow-up, exact time of performed blood test (preoperatively / perioperatively / postoperatively), laboratory analysis (e.g. immunoturbidimetric technique, EDTA as anticoagulation), etc. Statistical analyses will be performed using SPSS.

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Grading the quality of evidence

We will use the GRADE software to assess the quality of the included studies: bias risk; heterogeneity; publication bias; and other factors will be considered. The quality of the evidence will be graded using "very low", "low", "moderate", or "high".

Patient and Public Involvement

No patient involved.

DISCUSSION

One of the strengths of the proposed study is the application of a reproducible and transparent methods for the systematic review of the literature. In this protocol, we clearly define the types of studies, participants, interventions, comparators and outcomes to be included, together with the data sources, search strategy and data extraction and management to be used. By publishing this research protocol, we strengthen the clarity of the strategy and minimize the risk of bias. To the best of our knowledge, this is the first systematic review assessing the prognostic value of routine screening of IBP for surgical site infection after primary hip or knee arthroplasty. Although a few systematic reviews for diagnosis of periprosthetic joint infection were published in the past, their conclusions cannot be used for primary arthroplasty prognosis.[18–21] Therefore this review is solely focused on primary arthroplasty only. The findings may be applicable during preoperative measurements to assess the individual risk for surgical site infection to each patient, particularly if any discrepancies in antibiotic prophylaxis occur, because it needs to be individualized for each patient during mass operation, such as arthroplasty. This review will cover a large number of databases, including sources for a manual search. The use of articles in English language only is a limitation of the review. Poor-quality studies or reduced number of studies and heterogeneity in results may lead to difficulties in interpreting findings and to make a recommendation.

Registration and publishing

This systematic review protocol is registered on the International Prospective Register of Systematic Reviews (PROSPERO) with a trial number, CRD42020147925 (https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=147925) dated (28 April 2020). The findings of the review will be published in international, peer-reviewed journals.

Author contributions

Review conceptualization was designed by PD and ARP. Methodology of study design and literature search strategies were developed by PD, ARP and KML. Design of study quality risk assessment tools, data extraction, data synthesis and statistical tests were developed by PD, ARP and KML. Project supervision was carried out by JM. PD wrote the manuscript draft according to PRISMA-P. Draft revision was undertaken by all the authors.

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Competing interest

None declared.

Patient consent

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Not required.

Ethics approval

Not required.

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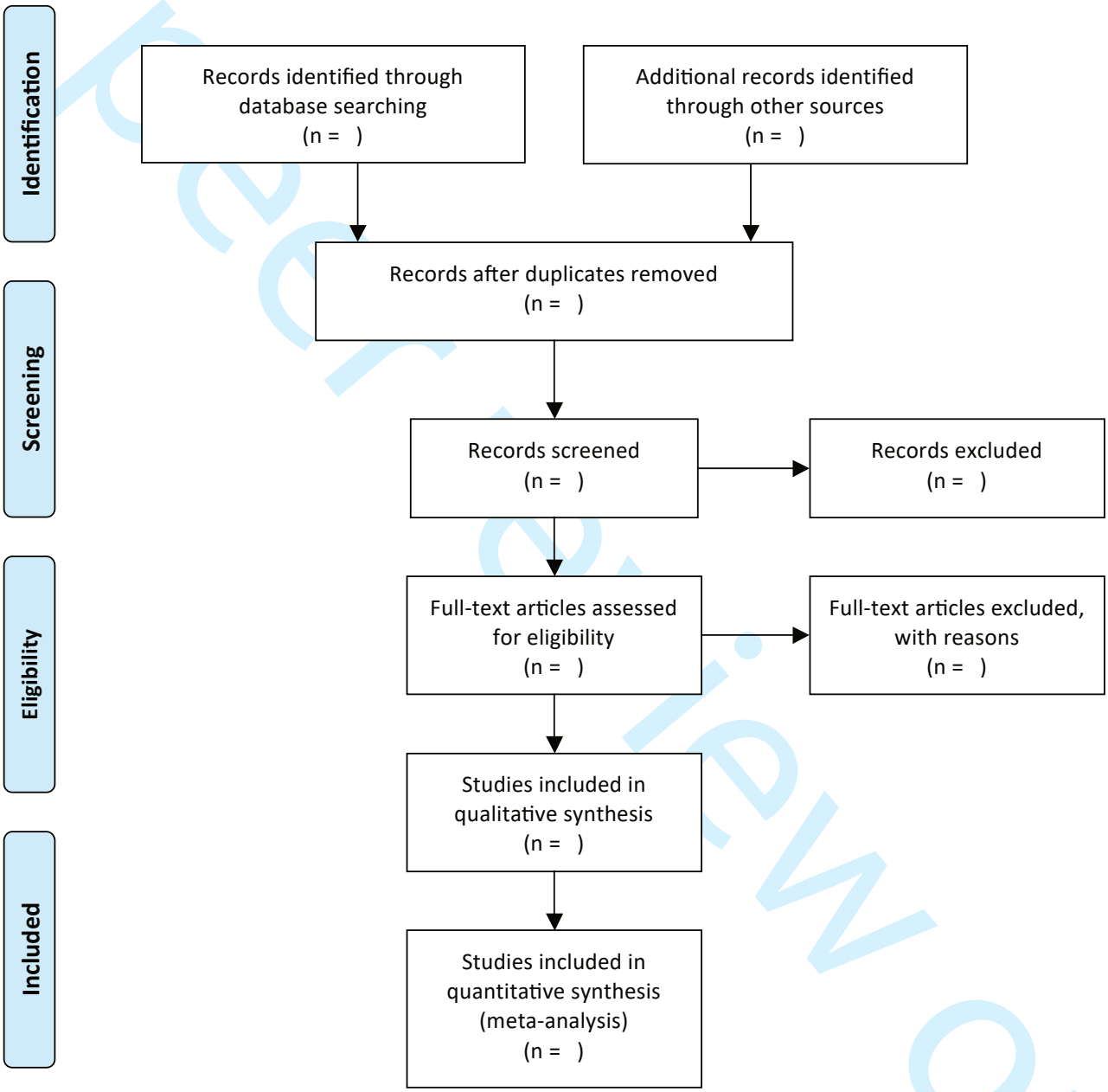
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Figure legend

Figure 1: The PRISMA flow chart

Legend: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	n/a
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	9
Amendments			

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	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	n/a
Support			
Sources	#5a	Indicate sources of financial or other support for the review	9
Sponsor	#5b	Provide name for the review funder and / or sponsor	9
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	9
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	4
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	5
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	5
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	5
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	6
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	6
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	6
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for	7

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		obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	8
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	5
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	8
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	8
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	8
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	8
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	8
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	8
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	8

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Inflammatory blood parameters as prognostic factors for surgical site infection after primary hip or knee arthroplasty: a systematic review protocol

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Inflammatory blood parameters as prognostic factors for surgical site infection after primary hip or knee arthroplasty: a systematic review protocol.

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Abstract

Introduction

Surgical site infection (SSI) is a potential complication of surgical procedure. SSI after implant surgery is a disaster both for patients and surgeons. Although predictive tools for SSI are available, none of them estimate early infection based on inflammatory blood parameters. The inflammatory process can be measured using several parameters including interleukin-6, C-reactive protein, neutrophil to lymphocyte ratio, white blood cell count, erythrocyte sedimentation rate, or procalcitonin. This systematic review aims to determine whether inflammatory blood parameters could be used as significant predictive factors for SSI after primary hip or knee arthroplasty.

Methods and analysis

A systematic review of randomized controlled trials, cross-sectional studies, case-control studies, and cohort studies, published in English, will be searched in the following electronic bibliographic databases: MEDLINE, Embase, PubMed, Cochrane Central Register of Controlled Trials, and Web of Science. Studies performed in adult patients of all ages who underwent knee or hip arthroplasty, studies containing data on the risk/prognostic factors for pre or post knee or hip arthroplasty SSI and studies with a minimum follow up of 30 days after surgery will be included. A standardized form will be used to extract data from the included studies comprising study characteristics, participant characteristics, details of the intervention, study methodology and outcomes. Quality Assessment of Diagnostic Accuracy tool, 2nd version, and Standards for Reporting of Diagnostic Accuracy Studies checklist will be used to assess risk of bias. Heterogeneity will be assessed using Cochran χ^2 statistic and I^2 statistics where applicable. Grading of Recommendations Assessment, Development and Evaluation and Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidance will be used to report findings.

Ethics and dissemination

No ethics approval is required. The findings will be disseminated at national and international scientific sessions, also to be published in a peer-reviewed journal.

PROSPERO registration number

CRD42020147925

Keywords

Adult orthopaedics, hip, knee, infection control

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Strengths and limitations of this study

The protocol reduces the feasibility of duplication, clarifies the methods and processes used, reduces potential review authors' biases and allows peer review of planned methods. The protocol offers the highest level of evidence for informed decisions based on this systematic review.

To the best of our knowledge, this will be the first systematic review assessing the prognostic value of routine screening of inflammatory blood parameters for surgical site infection (SSI) in primary hip or knee arthroplasty.

The scarcity of studies focused on patients after primary knee or hip arthroplasty containing data on the risk/prognostic factors for SSI with a minimum follow up of 30 days, an expected heterogeneity in design, varying methodological quality across studies and limitation of review search to English language studies may hinder meta-analysis and interpretation of findings.

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Introduction

Surgical site infection (SSI) is a potential complication following surgical procedure.[1] The presence of SSI leads to an increase both in morbidity and in mortality rates, thus contributing to higher healthcare costs.[2,3] In orthopaedics, SSI after implant surgery is a disaster both for the patients and surgeons.[4] SSI is defined as an infection occurring up to 30 days after surgery (or up to one year after surgery in patients receiving implants) and affecting either the incision or deep tissue at the operation site.[5] SSI prevalence may be reduced by various preoperative and postoperative measures. One of the measurements in which SSI is affected is antibiotic prophylaxis.[6,7] It is desirable to adjust antibiotic prophylaxis length for arthroplasty, according to individual SSI risk estimated by appropriate processes. Although predictive tools for SSI, such as ACS (*American College of Surgeons*) surgical risk calculator, or NHSN (*The National Healthcare Safety Network*) risk index are available, none of them estimate early infection based on inflammatory blood parameters (IBP).[8,9] In the acute-phase, cells of the immune system migrate to the site of injury in a carefully orchestrated sequence of events mediated by cytokines and acute-phase proteins.[10] The inflammatory process can be measured applying one of the several biochemical markers, including interleukin-6 (IL-6), C-reactive protein (CRP), neutrophil to lymphocyte ratio (NLR), white blood cell count (WBC), erythrocyte sedimentation rate (ESR) or procalcitonin (PCT). NLR is an easy, cheap, widely used, robust and convenient biomarker during inflammation, and has been shown to predict outcomes in cardiovascular diseases.[11,12] After a surgical trauma neutrophil blood levels via cytokines are increased, and the lymphocyte count is decreased. ESR is the rate how quickly red blood cells settle at the bottom of test tube with blood sample. In case of disease, high ESR is commonly due to an increased protein level in the blood, such as the increase of acute-phase proteins in inflammatory diseases. The ESR is nonspecific and, therefore, of little diagnostic value, but serial tests enable to follow the progress of the disease more efficiently. CRP, also known as nonspecific acute-phase response marker, is instigated by tissue damage, infection, inflammation, and malignancy. CRP is under control by IL-6 originating at the site of pathology. Serum concentration starts to rise within 6 hours, peaking in around 48 hours, from the time the given pathology occurs. CRP is still the most widely used biomarker for diagnosing postoperative infections.[12,13] PCT is one of the several bloodstream biomarkers investigated as an early predictor of sepsis. It is produced rapidly in response to bacterial stimuli but not in systematic inflammatory response syndrome of non-infectious origin.[12] IL-6 is one of the polypeptides secreted by cells triggered in response to injury, in the broad category of small proteins cytokines. The role of cytokines is to organize a concerted response to tissue trauma, involving cells recruited to the injury site. IL-6 is classified as the inflammation cytokine of acute-phase. To better understand the clinically significant responses to surgical trauma, it is important to elucidate the underlying mechanisms that regulate the inflammatory response to this type of tissue injury. The native immune system recognizes bacteria through pattern recognition receptors, such as toll-like receptors. These receptors stimulate, among others, the secretion of IL-6, which is also stimulated by interleukin-1 (α/β) and the tumour necrosis factor (α/β).[11] The present systematic review will primarily assess the response of IBP to trauma caused by orthopaedic surgery, and evaluate the clinical utility of quantitative measurements of IBP as predictive factors for infection. To our knowledge, there is no published systematic review addressing the same question. Following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA-P) checklist[14] as guidance, we propose a systematic and reproducible strategy to query the available literature.

Methods and analysis

The search strategy will be conducted according to the available resources, promoting methodological transparency and improving the reproducibility of the results and evidence synthesis. Additionally,

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using the Population, Intervention, Comparison, Outcome and Study design (PICOS) strategy, we elaborated the guiding question of this review in order to ensure the systematic search of available literature: *Can IBP (IL-6, CRP, NLR, ESR, WBC, PCT) be effective as predictive factors for SSI in knee or hip arthroplasty?* The International Prospective Register of Systematic Reviews registration number is CRD42020147925. The planned period to compose this review study is from 1 September 2019 to 30 September 2021.

Inclusion criteria for study selection

A summary of the participants, interventions, comparators, and outcomes considered, as well as the type of studies was made according to PICOS strategy.

Types of studies

Cross-sectional, cohort, case-control, randomized controlled trial study designs will be included, limited to publications in English language. Case reports will be excluded.

Types of participants

Studies with adult patients of all ages who underwent primary hip or knee arthroplasty will be included. The minimum follow-up after the operation must be 30 days. Patients with clinical signs of infection, neoplasia, inflammatory diseases (rheumatoid arthritis, Crohn, HIV, etc.) will be excluded.

Types of interventions or exposures

IBP determination such as IL-6, CRP, WBC, ESR, NLR and PCT.

Types of compactors

We will compare the prognostic value of IBP to each other.

Types of outcomes measures

The outcome measures will be the IBP that can be used as significant predictive factors for SSI development. SSI is defined as an infection occurring up to 30 days after surgery (or up to one year after surgery in patients receiving implants) and affecting either the incision or deep tissue at the operation site[5]. Periprosthetic joint infection as subcategory to SSI will be defined according to the original diagnostic criteria proposed by the Musculoskeletal Infection Society (MSIS) Workgroup in 2011[15], the Infection Disease Society of America (IDSA) in 2013[16], the International Consensus Meeting (ICM) in 2013[17], and finally according to the new proposed definition from 2018 using an evidence-based and weight-adjusted scoring system.[18]

Search methods for the identification of studies

Electronic searches

We will conduct this systematic review and meta-analysis in accordance with the PRISMA guidelines.[19] The following electronic bibliographic databases will be searched: PubMed (MEDLINE), Cochrane Central Register of Controlled Trials (Cochrane databases), Web of Science. The search terms will include both medical subject headings (MeSH) and keywords related to the topics about IBP or risk factors, or prosthesis-related infection or SSI and knee or hip arthroplasty. The sample of the search strategy for MEDLINE is presented in Table 1. Similar search strategy will also be utilized to any other electronic database.

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Table 1 Search strategy for MEDLINE database

Number	Search terms
1	prosthesis joint infection or prosthesis related infection
2	prosthetic infection or inflammation or acute-phase reaction or seroma or abscess or systematic inflammatory response syndrome or sepsis
3	wound infection
4	surgical site infection or surgical wound infection
5	arthroplasty or arthroplasty, replacement or arthroplasty replacement, hip or arthroplasty replacement, knee
6	joint replacement
7	joint prosthesis or hip prosthesis or knee prosthesis or metal-on-metal join prostheses
8	joint arthroplasty
9	risk factor or risk factors
10	predictor
11	inflammatory marker
12	biological markers or hematologic tests
13	leukocyte count
14	blood sedimentation rate
15	neutrophils
16	lymphocytes
17	neutrophil to lymphocyte ratio
18	interleukin 6
19	c reactive protein
20	procalcitonin
21	1 or 2 or 3 or 4
22	5 or 6 or 7 or 8
23	9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20
24	21 and 22 and 23
25	limit to 24 (english language and yr="1902-Current")

Note: For purpose of this table, search terms are presented without additional settings like “focus, explode, multi-purpose”, etc.

Searching other resources

Manual searches will be performed mainly to retrieve dissertations, ongoing trials, internal reports etc, in order not to miss any relevant studies. Additionally, we will search the bibliographies of included publications manually for further, potentially eligible papers.

Data collection and analysis

Selection of studies

All electronically and manually searched records will be merged to remove duplicate citations. Titles, abstracts and full-texts of studies retrieved using the search strategy will be screened independently through the EndNote software by at least two review authors (PD, ARP) to identify studies that potentially meet the inclusion criteria in the two-stage selection process. The initial screening of studies will be based on the information contained in their titles and abstracts, to be conducted by two independent investigators (PD and ARP). In case reviewers disagree, the article will be re-evaluated and if the disagreement persists, a third reviewer (KM) will be asked to make the final decision. Full-paper screening will be conducted by the same independent investigators. The reasons for literature

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excluding will be recorded. The details of the selection process are shown in the PRISMA flow chart (Fig. 1).

Data extraction and management

A standardized form will be used to extract data from the included studies for study quality assessment (Grading of Recommendations Assessment, Development and Evaluation (GRADE) algorithm)[20] and evidence synthesis. Extracted information will include study characteristics (design, settings, year of publication), participant characteristics (mean age, percentage of each gender), details of intervention (IBP with all characteristics), study methodology and outcomes (SSI, prosthesis-related infection, risk/prognostic factors, limitations). Missing data will be discussed with the authors of studies, where applicable. Two independent Microsoft Excel spreadsheets will be elaborated for two reviewers (PD and ARP) to summarize the data from the included studies. Then, the spreadsheets will be combined into one. Disagreements will be resolved by a third investigator (KM).

Assessment of risk of bias in included studies.

Quality Assessment of Diagnostic Accuracy (QUADAS-2) tool[21] and Standards for Reporting of Diagnostic Accuracy Studies (STARD) checklist[22] will be used to assess the risk of bias in the included studies by two reviewers (PD and ARP). The disagreement in the risk of bias assessment or the level of evidence (GRADE) will be resolved by consensus.

Dealing with missing data

If the required data are unclear or not reported in any of the articles, the reviewer will contact the author by email, or mail to collect the missing data as completely as possible.

Assessment of heterogeneity

Heterogeneity will be assessed using Cochran χ^2 statistic and I^2 statistics where applicable.

Data synthesis and analysis

We will conduct a narrative synthesis for the findings from the included studies, structured around the type of intervention, target population characteristics, type of outcome and intervention content. One reviewer (PD) will synthesize the data. Three other reviewers will check the results and discuss any discrepancies to reach consensus. If there is substantial clinical homogeneity of the participants and assessment methods, meta-analysis will be conducted to estimate the risk/protectability of each IBP in relation to knee or hip arthroplasty.

Assessment of publication bias

The funnel plot will be used to judge whether a publication bias exists if at least 10 articles are included. We will investigate the (a)symmetry in (inverted) funnel plots which show the association between effect size and the accuracy of the study.

Subgroup analysis

When the assessment suggests significant heterogeneity, we will perform a subgroup analysis based on study-level characteristics including diagnosis of the knee or hip osteoarthritis, type of surgery (based on how invasive method was used), duration of surgery, use of antibiotic prophylaxis (with other characteristics including dose and timing), thrombosis prevention, average duration of follow-up, exact time of performed blood test (preoperatively / perioperatively / postoperatively), laboratory analysis (e.g. immunoturbidimetric technique, ethylenediaminetetraacetic acid as anticoagulation),

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American Society of Anaesthesiologists physical status classification system etc. Statistical analyses will be performed using SPSS.

Grading the quality of evidence

We will use the GRADE software to assess the quality of the included studies: bias risk; heterogeneity; publication bias; and other factors will be considered. The quality of the evidence will be graded using "very low", "low", "moderate", or "high".

Patient and Public Involvement

No patient will be involved.

DISCUSSION

The reproducible and transparent procedure for the systematic review of the literature is one of the strengths of our proposed protocol. We precisely described the types of studies, participants, interventions, comparators and outcomes to be included, as well as the data sources, search strategy, data extraction processes, data analysis and synthesis, and assessment of risk of bias. By publishing the research protocol, we reinforce the clarity of the strategy and minimize the risk of bias, namely selective outcome reporting. To the best of our knowledge, this is the first systematic review assessing the prognostic value of routine screening of IBP for SSI after primary hip or knee arthroplasty. Although a few systematic reviews for diagnosis of periprosthetic joint infection were published in the past, their conclusions cannot be used for primary arthroplasty prognosis.[23–26] Therefore this review is solely focused on primary arthroplasty only. The findings may be applicable during preoperative measurements to assess the individual risk for SSI to each patient, particularly if any discrepancies in antibiotic prophylaxis occur, because it needs to be individualized for each patient during mass operation, such as arthroplasty. This review will cover a large number of databases, including sources for a manual search. The use of articles in English language only is a limitation of the review. Poor-quality studies or reduced number of studies and heterogeneity in results may lead to difficulties in interpreting findings and to make a recommendation.

Registration and publishing

This systematic review protocol is registered on the International Prospective Register of Systematic Reviews (PROSPERO) with a trial number, CRD42020147925 (https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=147925) dated (28 April 2020). The findings of the review will be published in international, peer-reviewed journals.

Author contributions

Review conceptualization was designed by PD and ARP. Methodology of study design and literature search strategies were developed by PD, ARP and KML. Design of study quality risk assessment tools, data extraction, data synthesis and statistical tests were developed by PD, ARP and KML. Project supervision was carried out by JM. PD wrote the manuscript draft according to PRISMA-P. Draft revision was undertaken by all the authors.

Funding statement

PD as a PhD student is supported by Charles University (Project SVV 260 551).

Competing interest

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None declared.

Ethics and dissemination

No ethics approval is required. The findings will be disseminated at national and international scientific sessions, also to be published in a peer-reviewed journal.

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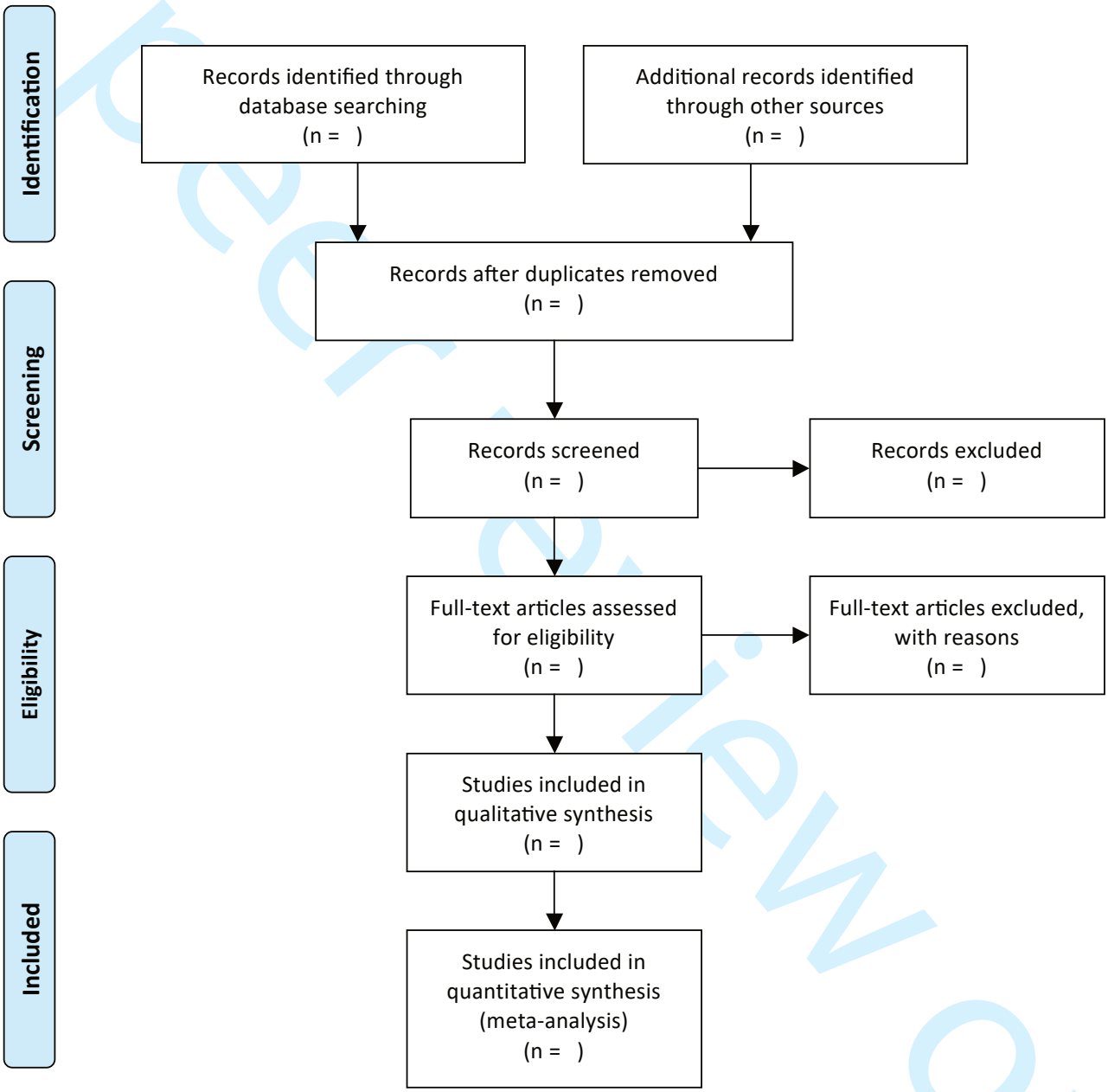
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Figure legend

Figure 1: The PRISMA flow chart

Legend: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Reporting checklist for protocol of a systematic review.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

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Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Syst Rev. 2015;4(1):1.

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		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	n/a

Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	8
Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	n/a
Support			
Sources	#5a	Indicate sources of financial or other support for the review	8
Sponsor	#5b	Provide name for the review funder and / or sponsor	8
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	8
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is	4

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		already known	
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	4
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	4
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	4
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	6
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	6
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	6
Study records - data collection	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any	7

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process		processes for obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	7
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	5
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	7
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	7
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	7
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	7
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	7
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within	7

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		studies)	
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	8

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